

Weather-Based Crop Advisory System

Requirement Analysis Report

Version 1.0

Academic Project Documentation

1. Introduction

Agriculture is strongly influenced by weather conditions such as rainfall, temperature, humidity and wind. A general weather forecast provides useful information, but it does not directly explain what action should be taken for a particular crop. The Weather-Based Crop Advisory System is proposed as a simple decision-support application that connects weather information with crop information and provides understandable recommendations.

2. Problem Statement

Farm activities such as irrigation, spraying and other field operations can be affected by short-term weather changes. Farmers may need to interpret several weather parameters manually before deciding what to do. The proposed system addresses this gap by using location, crop and crop-stage information together with weather conditions to generate relevant advisory messages.

3. Existing System

- Weather forecasts and agricultural information are often checked separately.
- General weather applications normally do not provide crop-specific recommendations.
- Manual interpretation of several weather parameters can be difficult.
- Important changes in forecast conditions may be missed.

4. Proposed System

The proposed system allows a farmer to register/login, select a farming location and crop, retrieve weather information and view crop-specific advisories. An advisory engine evaluates active rules against available weather conditions. An administrator can maintain crop information and advisory rules.

5. Objectives

- Provide weather information for a selected farming location.
- Allow selection of crop and crop stage where applicable.
- Generate actionable crop-specific recommendations.
- Display alerts for weather conditions that may affect farm activities.
- Provide a simple interface suitable for non-technical users.

6. Scope

The system covers authentication, location selection, crop selection, weather retrieval, rule-based advisory generation, advisory display and administrative management of crops and rules. It is a decision-support tool and does not guarantee crop yield, replace professional agricultural advice or directly control farm equipment.

7. Stakeholders

Stakeholder	Role
Farmer	Uses weather data and crop advisories for farming decisions.
Administrator	Maintains crop data and advisory rules.
Developer	Develops, tests and maintains the software.
Weather Data Provider	Supplies external weather observations and forecasts.
Agricultural Expert (optional)	Can review the agricultural logic used in advisory rules.

8. Functional Requirements

ID	Requirement
FR-01	The system shall allow users to register and log in.
FR-02	The system shall allow a farmer to select a farming location.
FR-03	The system shall allow selection of a supported crop.
FR-04	The system shall accept crop-stage information where applicable.
FR-05	The system shall retrieve weather information for the selected location.
FR-06	The system shall evaluate active advisory rules.
FR-07	The system shall generate crop-specific recommendations.
FR-08	The system shall display weather-related alerts.
FR-09	The administrator shall manage crop information.
FR-10	The administrator shall manage advisory rules.

9. Non-Functional Requirements

Category	Requirement
Usability	Clear labels, simple navigation and readable advisory messages.
Performance	Normal operations should respond within a reasonable time for expected project usage.
Security	Authentication and role-based authorization shall protect restricted functions.
Reliability	Unavailable weather data shall not be treated as valid input.
Maintainability	Weather integration and advisory logic should remain modular.
Scalability	Additional crops and rules should be addable later.

10. Advisory Rule Table

Rule ID	Condition	Recommended Action	Priority
R01	High probability of rain	Consider postponing irrigation when immediate watering is not required.	High
R02	Unsuitable or strong wind	Avoid pesticide or fertilizer spraying during unsuitable wind conditions.	High
R03	Very high temperature	Check soil moisture and irrigation requirements more frequently.	Medium
R04	High humidity with wet conditions	Increase attention to weather-related crop risk.	Medium
R05	Weather data unavailable	Do not generate weather-based advice; display a data-unavailable message.	High

11. Data Flow Diagram – Level 0

The context-level DFD shows the main external entities and the Weather-Based Crop Advisory System as a single process.

DFD diagrams are included in the separate UML-and-DFD document and Appendix B of the SRS.

12. Data Flow Diagram – Level 1

The Level 1 DFD decomposes the system into authentication, weather processing and advisory generation, with data stores for users, crops/rules and generated advisories.

13. Data Requirements

Entity	Key Information
User	User ID, name, login details, password hash and role.

Entity	Key Information
Crop	Crop ID, crop name, supported stages and status.
Location	Location ID, name and coordinates/location identifier.
Weather Data	Timestamp, location, temperature, precipitation/rain, humidity, wind and forecast values.
Advisory Rule	Rule ID, crop/crop stage, condition, message, severity and status.
Advisory	Generated message, rule reference, user reference and timestamp.

14. Feasibility Analysis

Technical: The project can be implemented with commonly available software technologies and an external weather API. Rule-based advisory logic is manageable for an academic prototype.

Economic: The prototype can use free/open-source development tools and an appropriate weather-service plan.

Operational: The workflow is simple: select location, select crop, view weather and read the recommendation.

15. Assumptions and Constraints

- The weather service provides usable information for the selected location.
- The farmer provides correct crop and location information.
- Advisory rules are reviewed before activation.
- Forecasts can change and should not be treated as guarantees.

16. Acceptance Criteria

1. A registered user can log in.
2. A farmer can select a location and crop.
3. Weather information is displayed when the external service is available.
4. A configured weather condition produces the corresponding advisory.
5. The system displays a clear message when weather data is unavailable.
6. Unauthorized users cannot perform administrator-only operations.

17. Summary

The requirement analysis defines a focused decision-support system that converts weather conditions into understandable crop advisories. The scope is suitable for an academic project and can be extended with more crops, regional rules, notifications and additional agricultural data.